



All-in-one chip for smartphones and connected electronics

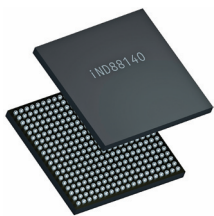
STMicroelectronics has introduced the ST54M, a secure mobile chip that combines post-quantum cryptography (PQC), NFC, secure element, and eSIM functions in one device. It is designed for smartphones and connected electronics to support applications such as contactless payments, digital identity, transit tickets, access control, driving licences, digital car keys, and connectivity services. It also offers more memory and an RF front end for wireless communication, compact antenna designs, and single-ended configurations, making it suitable for applications such as mobile payment terminals and wireless charging systems. The ST54M targets industries including mobile networks, banking, government, transportation, automotive, and digital services that require long-term security for connected devices.



STMicroelectronics
<https://www.st.com/content>

Edge AI SoC for robotic and automotive cameras

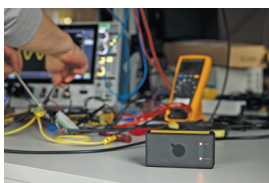
Indie Semiconductor has launched the iND881, an edge AI system-on-chip (SoC) designed to power smart cameras for automotive and robotic applications. The device combines an AI compute engine with Indie's low-latency multi-camera image signal processor to support real-time perception systems. The SoC targets automotive applications such as driver monitoring systems, occupant monitoring systems, smart eMirrors, and blind-spot detection. It is also designed for robotics applications, including autonomous mobile robots and humanoid robots, where multiple sensors are used for navigation and



awareness. The iND881 includes a low-latency H.264 encoder for multi-channel video compression and streaming.
Indie Semiconductor
<https://www.indie.inc>

3D sensor designed to improve human-robot safety

Sonair's ADAR One 3D sensor is designed to improve human-robot safety by helping robots detect people and obstacles across a 3D space. The sensor uses acoustic detection and ranging (ADAR) technology instead of light-based sensing methods used in cameras and traditional lidar systems. The sensor is designed for autonomous mobile robots, industrial automation systems, and humanoid robots, and can be integrated into compact robot designs. It works with cameras, AI systems, and motion-control platforms by providing an independent way to monitor the surrounding environment.



Sonair
<https://www.sonair.com>

Ultra-thin LED cable modules for single-use endoscopes

OMNIVISION has introduced OVMed medical-grade LED cable modules for single-use endoscopes. The ready-to-integrate modules combine a Camera-CubeChip image sensor, wafer-level optics, LED illumination, and cabling into one package, helping endoscope manufacturers reduce development time and bring products to market faster. The thin cable is designed for small-diameter endoscopes, helping them reach deeper and narrower areas of the body. The modules are designed for medical endoscopes, including ophthalmoscopes, nasopharyngoscopes,



bronchoscopes, angioscopes, nephroscopies, ureteroscopes, choledochoscopes, spinoscopes, duodenoscopes, gastroscopies, laparoscopes, proctoscopes, cystoscopes, hysteroscopes, utero-renaloscopes, colonoscopes, and arthroscopies.
OMNIVISION
<https://www.ovt.com>

4nm driving chip for intelligent driving

BYD has introduced the Xuanji A3, which it says is China's first automotive-grade 4nm driving chip for intelligent driving. The chip delivers 700 TOPS of AI computing performance for advanced driver-assistance and autonomous driving tasks and has entered mass production. According to the company, it uses about 20% less power than comparable automotive AI chips.



BYD
<https://www.byd.com/in>

USB-C dock that allows change of functions

DockFrame is an open source, modular USB-C dock that supports framework-compatible expansion cards and custom tool cards, allowing users to change functions without replacing the dock. It features four hot-swappable expansion slots, enabling users to configure the dock with the ports and tools they need instead of using a fixed-function hub. Tool Cards can also work directly with USB-C devices, extending their use beyond the dock. DockFrame supports USB-C connectivity, power delivery, data transfer, and display output, while its LEGO-compatible enclosure allows users to stack multiple DockFrames or attach LEGO accessories.



DockFrame
<https://dockframe.com>